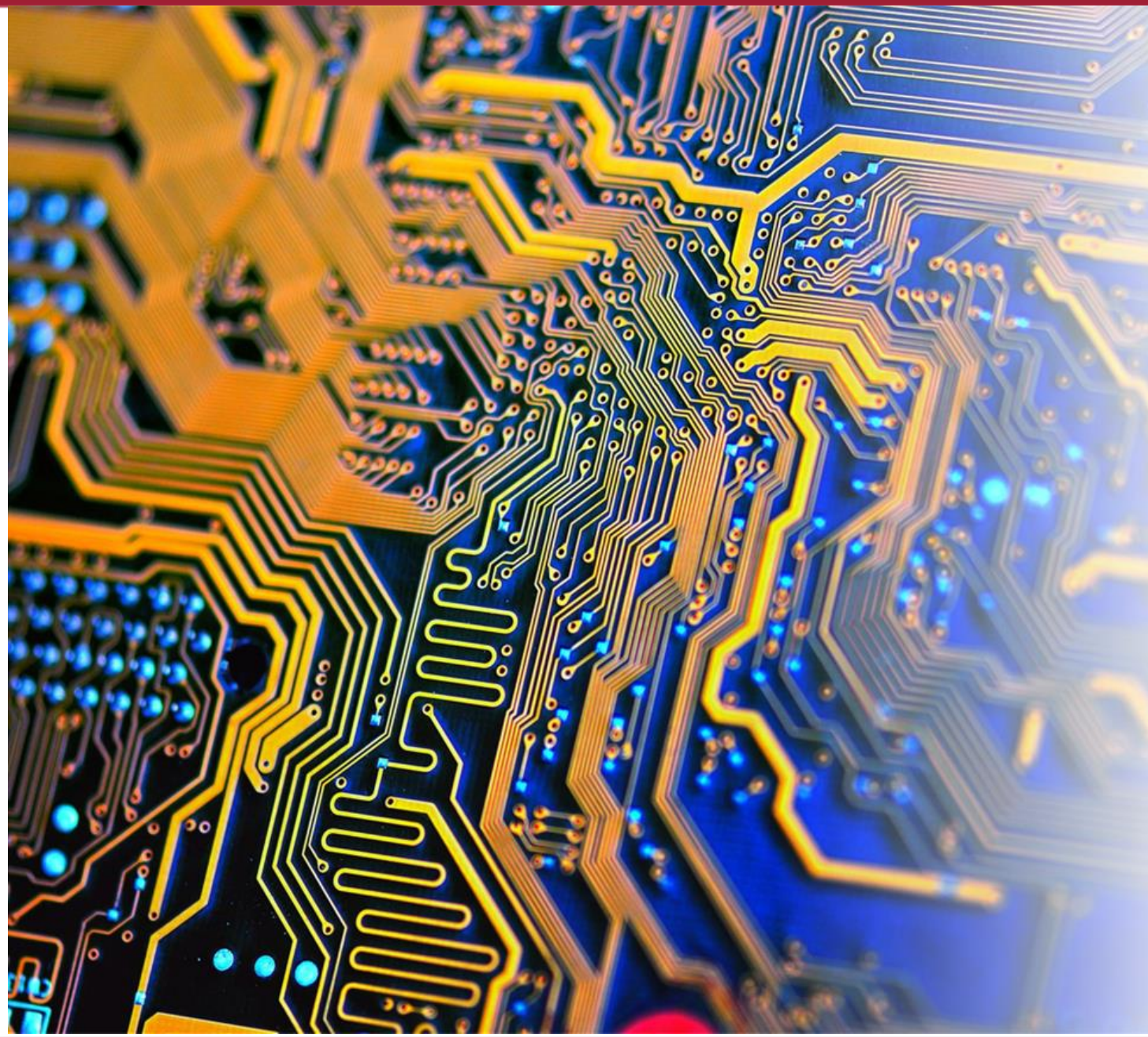


Designing Artificial Intelligence Products

Module 4: Designing Artificial Machines to Solve Problems

October 30, 2024

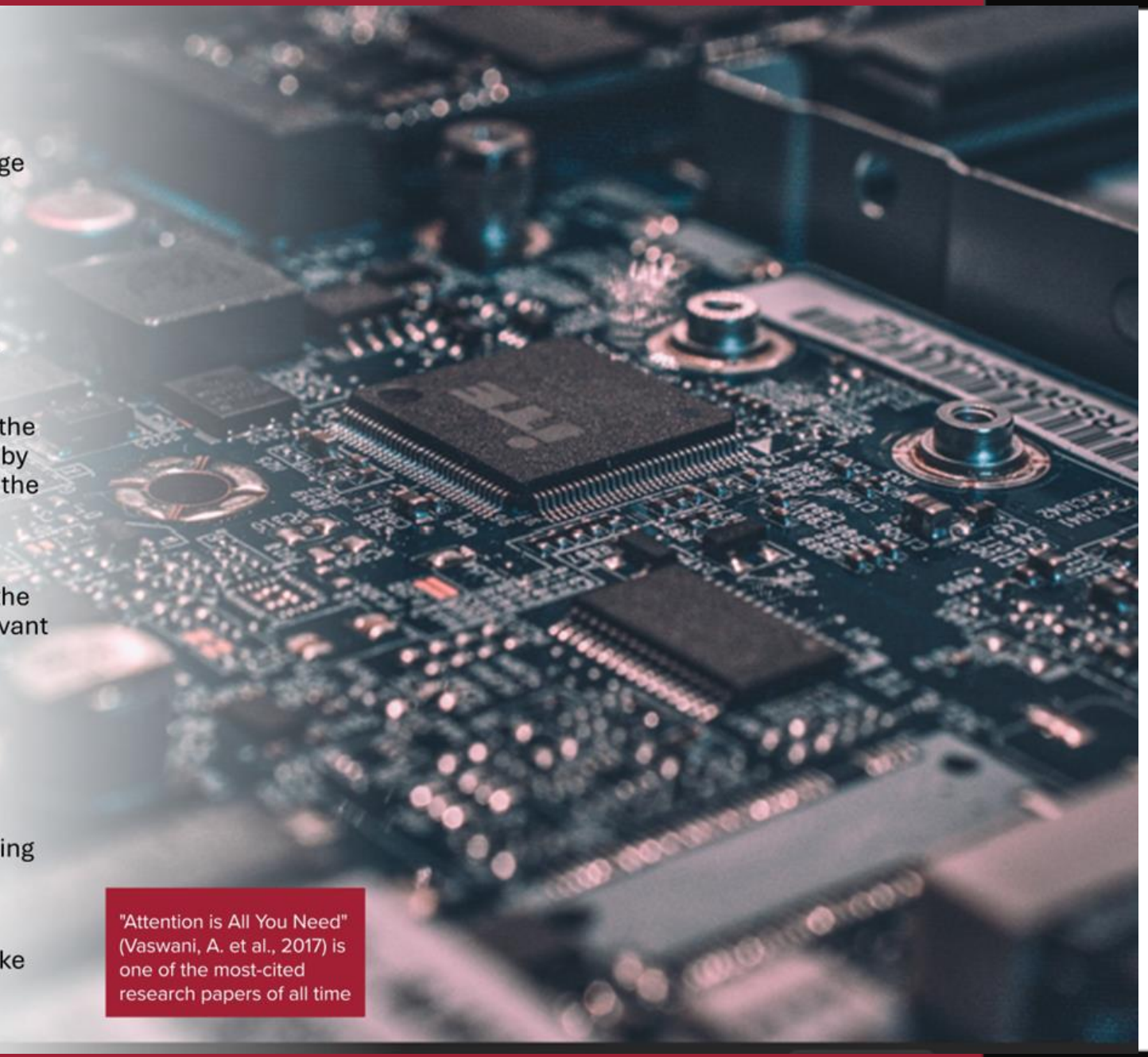
Amanda Fetch



Learning Outcomes

- Identify the superhuman intelligence used in an AI product or process
- Appraise the difficulties an AI technology can face
- Identify the key features and purpose of residual networks
- Describe how to augment a dataset using biomarkers
- Discuss the approvals necessary for human experimentation
- Discuss the implications of crowdsourcing data strategies
- Demonstrate how AI can solve healthcare problems

- Transformers are the foundational technology behind large language models (LLMs)
  - Architecture
    - Tokenizer
      - Breaks down text into smaller units (tokens) to be processed by the model
    - Encoder
      - Processes the input tokens, capturing the context and meaning of the input data by generating embeddings that represent the information
    - Decoder
      - Uses the processed information from the encoder to generate coherent and relevant output, forming the final response
- GPT-1 was the first major transformer model to hit the market
- LLMs have expanded to multimodal generative AI, including text, sound, image, and video
- These technologies are reshaping research institutions like MIT



"Attention is All You Need" (Vaswani, A. et al., 2017) is one of the most-cited research papers of all time

Stage 1: Figure – Ground and AI Categorization

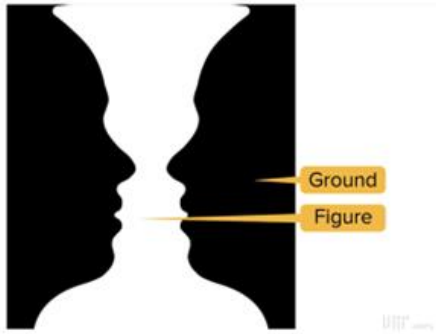


Figure 1

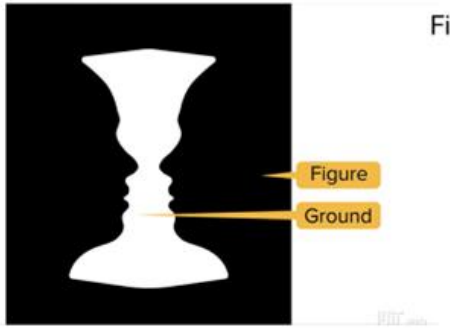


Figure 2

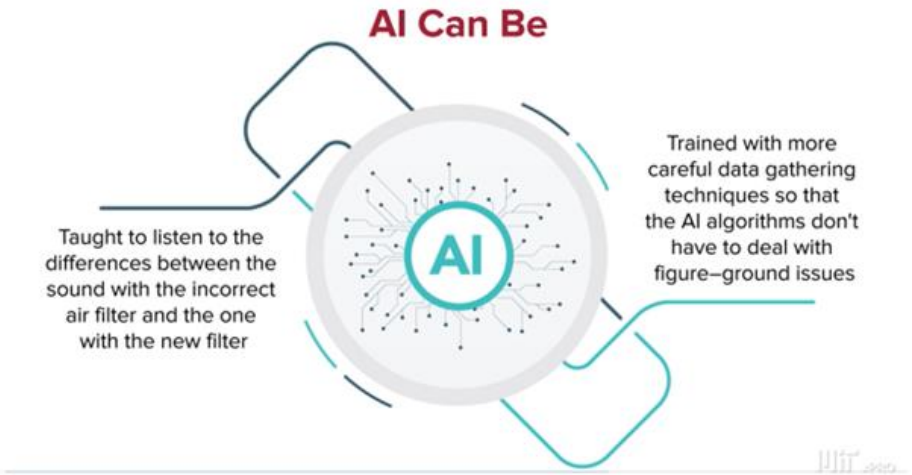
- Figure-ground segregation is essential for both human and machine-based visual recognition
- It involves identifying a figure (object of focus) from its background (ground)
- The first image illustrates how perception shifts between seeing a vase (white figure) and two faces in profile (black figure) depending on focus



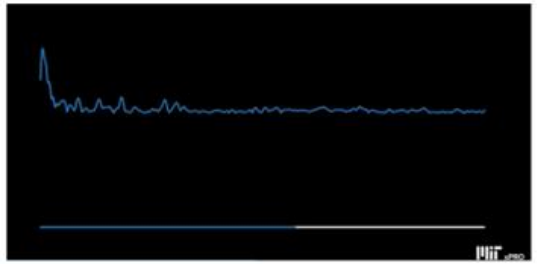
Figure-ground segregation extends beyond vision to sound processing, where AI distinguishes a specific sound (figure) from background noise

Applications include medical AI, such as analyzing a person's voice to detect early signs of conditions like Alzheimer's, dementia, or COVID

The figure-ground concept enables AI to isolate important data (figure) from irrelevant information (ground), whether in vision or sound analysis





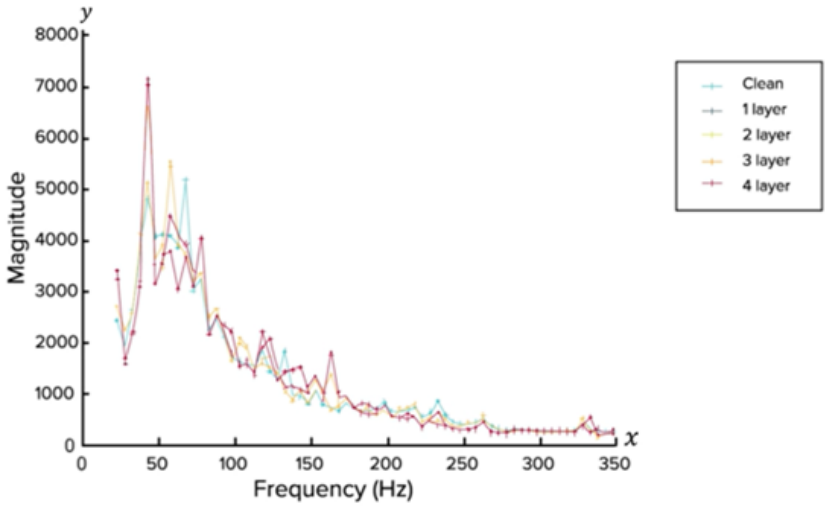


- Mechanics diagnose issues through sounds from key engine parts, like the valve train, timing chain, crankshaft bearing, and more
- AI can differentiate subtle differences in car sounds, such as identifying a dirty air filter, enabling accurate sound-based diagnostics
- Fourier transform sound analysis predicts when an air filter needs replacement, based on sound patterns
- "Shazam for Cars" Concept: AI-powered apps, like Shazam for music, could analyze vehicle sounds to diagnose problems, suggest fixes, and even order replacement parts or services automatically



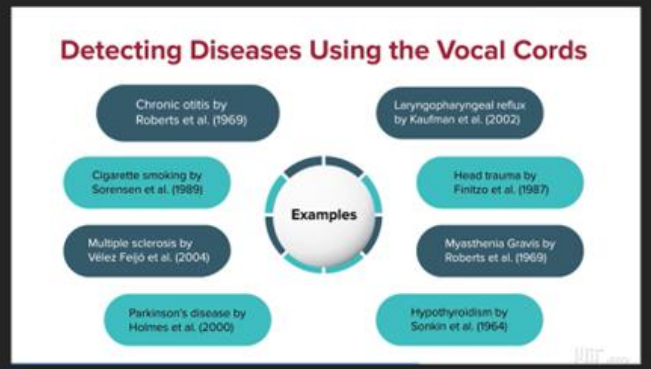
- MIT AutoID Lab researchers used AI to detect air filter cleanliness through sound frequency analysis
- The AI model can quantify dirt levels and predict filter replacement timing
- The research suggests that smartphones could use microphones for real-time air filter maintenance insights
- Related research used smartphone accelerometers to detect vehicle wheel imbalances with 90% accuracy
- These innovations suggest future potential for smartphones to diagnose mechanical issues by analyzing sound patterns.



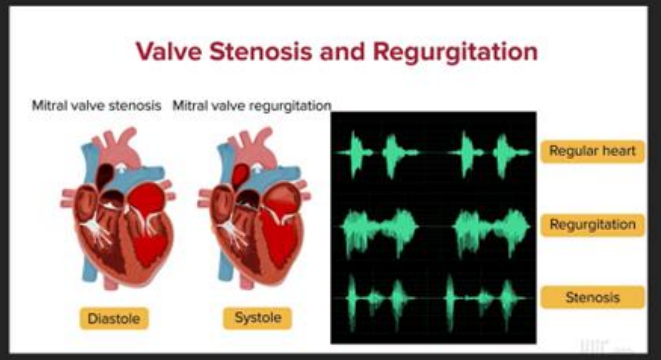


Source: *Siegel Joshua, Bhattacharya Rahu, Kumar Sumeet and Sharma Sanjay. "Air Filter Particulate Loading Detection Using Smartphone Audio and Optimized Ensemble Classification," Elsevier 66, (November 2017): 104-112*

- Fourier transform shows sound frequency changes based on air filter dirt levels
- Y-axis: Sound magnitude, X-axis: Frequency (Hz)
- Different colors represent cleanliness states (clean to dirty)
- Dirtier filters shift frequencies, reduce peaks
- AI uses this data for real-time maintenance prediction via smartphones



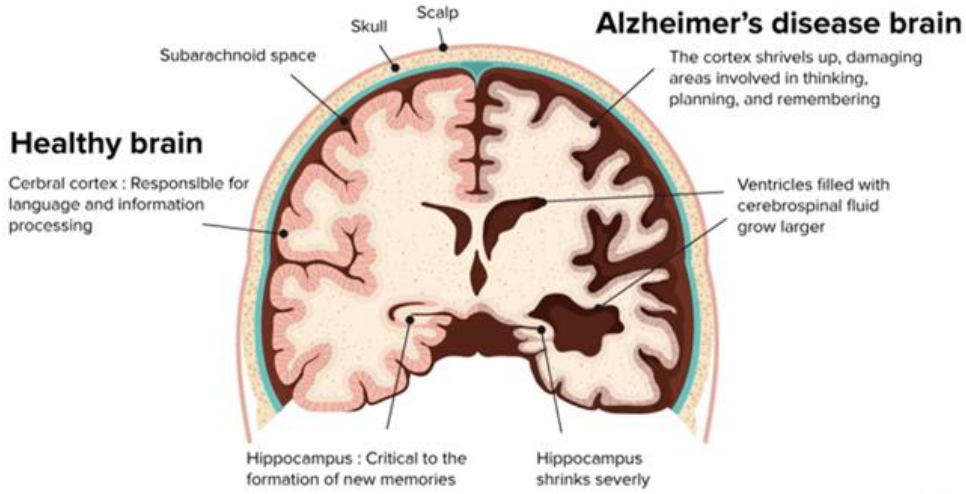




Stage I: Figure–Ground and AI Categorization — Digital Health



Source: Kokkinakis, Dimitrios, Kristina Lundholm Fors, Eva Björknes, and Arto Nordlund. "Data Collection from Persons with Mild Forms of Cognitive Impairment and Healthy Controls - Infrastructure for Classification and Prediction of Dementia. 2017



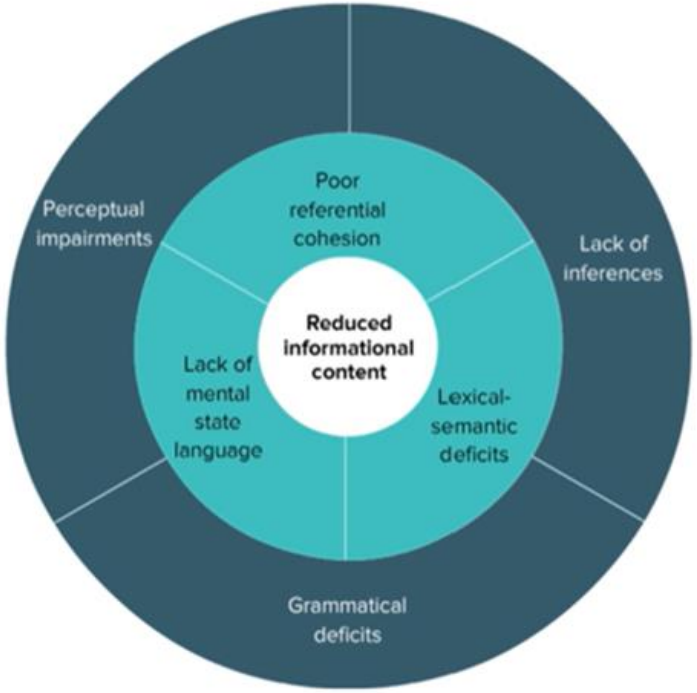
Stage 1: Figure – Ground and AI Categorization – Alzheimer's Example

- The comparison between healthy and Alzheimer's-affected brains highlights how structural changes lead to cognitive impairments
- Cognitive impairments can be identified through figure-ground tasks, like describing the "Cookie Theft" image
- In AI categorization, similar figure-ground techniques detect patterns and anomalies in data, automating the identification of cognitive-linguistic deficits linked to Alzheimer's
- AI can assist in early diagnosis by categorizing impairments based on patterns that align with the progression of Alzheimer's disease

NLP Techniques and Cognitive Impairment Identification

AI can use natural language processing techniques to:

- Identify the possible indicators of cognitive decay
- Establish a dementia scale
  - Identify the onset and progression of the disease
  - Identify finer metrics associated with each type of dementia













Stages of the AI Design Process

Stage one



Intelligence

Performance metrics



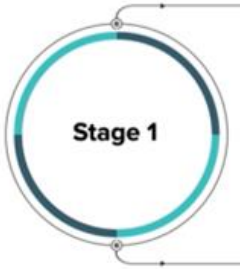
What are the target intelligence performance metrics that we want to implement with AI?

Scope



Understand the scope of what we're asking AI to achieve

Stages in Alzheimer's Diagnosis



Discriminating between Alzheimer's positive and negative subjects with an accuracy of over 99% and a scale tracking disease progression

Diagnosing Alzheimer's based on the description of subjects of the Cookie Theft scene

Using speech and mood changes for diagnosis

Defining the scope of the disease

AI Products and Their Superhuman Intelligence [Crowdsource Activity 4.1]

Aug 23 at 2:05pm

All Sections



Learning Outcome Addressed

1. Identify the superhuman intelligence used in an AI product or process.

This is a self-study activity and does not count toward the completion of the program.

Introduction

So far in this module, you have learned how AI can distinguish sounds and analyze language. In some contexts AI can analyze or create images, sounds, and language much faster and better than humans, some people say that it then exhibits superhuman intelligence. In this activity, you'll find a new application of this intelligence and share it with your fellow participants.

Task

1. Locate an article, a podcast, a video, or another form of media about a superhuman intelligence that may be use in an AI product or process.
2. Use Google Scholar to find a scientific article about the AI product you located.
3. Summarize your research by highlighting the pros and cons of the technology.
4. Using the four steps of the AI development process model, describe how you would design your chosen application.

Read the statements posted by your peers. Feel free to respond to your peers with thoughtful comments and questions to deepen the discussion.

Submission Instructions

- The suggested word count for this discussion is 150-200 words.
- The average completion time for this activity is 30 minutes.

Who Does Which Cough Belong to? [Drag-and-Drop Activity 4.1]



Learning Outcome Addressed

2. Appraise the difficulties an AI technology can face.

This is a self-study activity and does not count toward the completion of the program.

Context

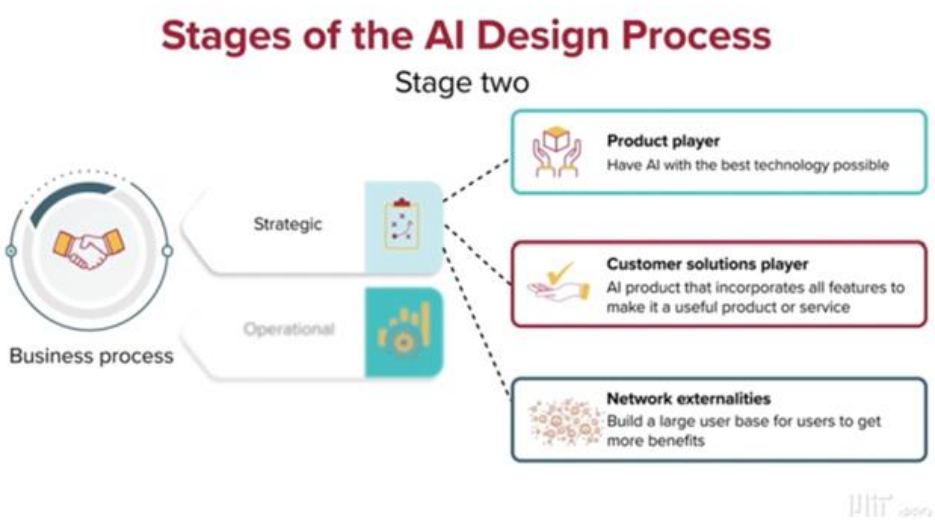
So far in this module, you have seen how to design AI intelligence to record (i.e., listen to) sounds and categorize them. In this exercise, see whether you can distinguish between the sounds of different coughs.

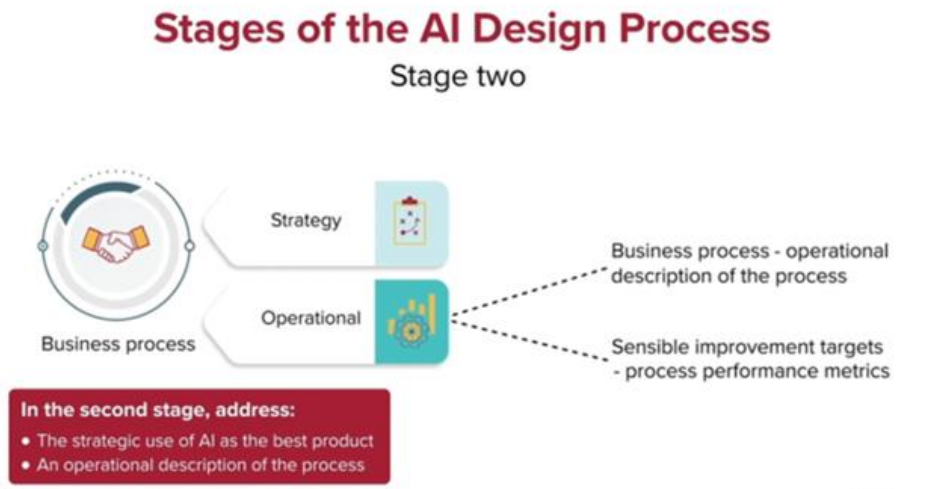
Task

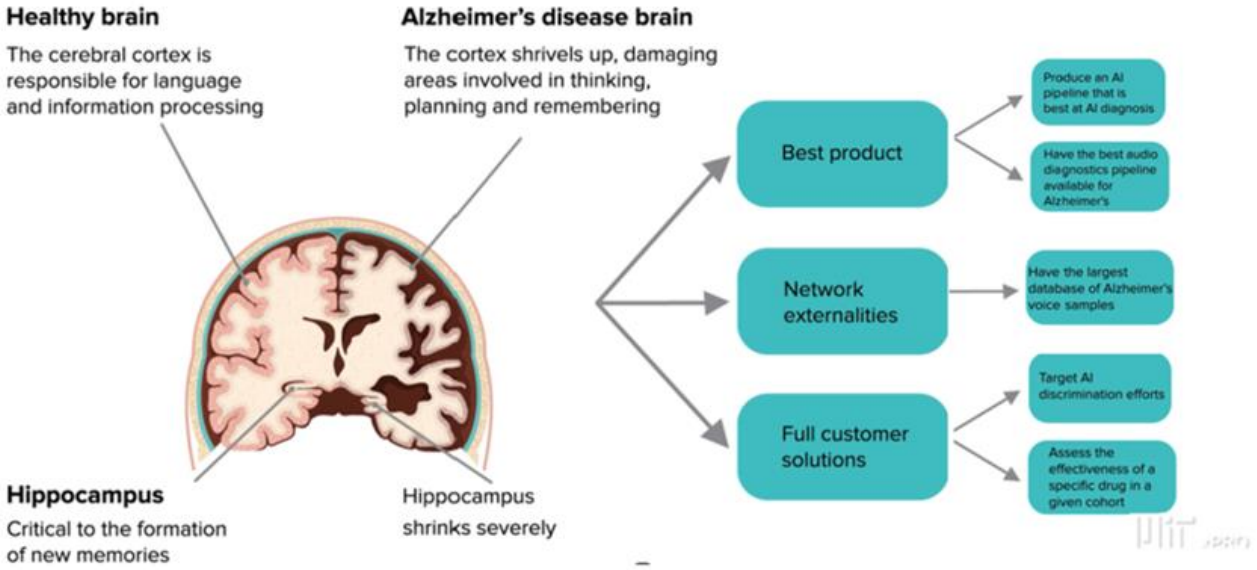
1. Listen to the cough recordings provided below.
2. Discern which of these coughs belongs to a child, an adult female, and an adult male.

The average completion time for this activity is 10 minutes.

Stage II: Strategy and Operations — Alzheimer's

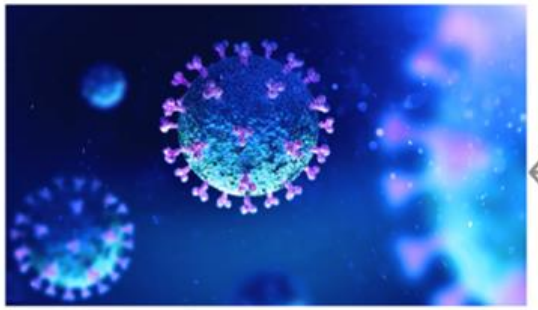


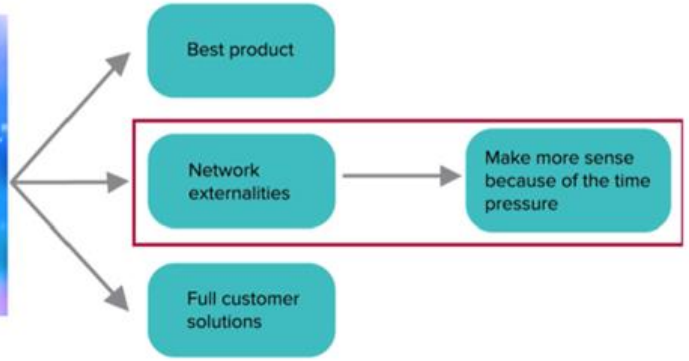




Stage II: Strategy and Operations — COVID

What are some examples of industries or technologies where being the first to market can create a lasting competitive advantage, similar to how network externalities played a critical role in AI diagnostic tools during COVID-19?

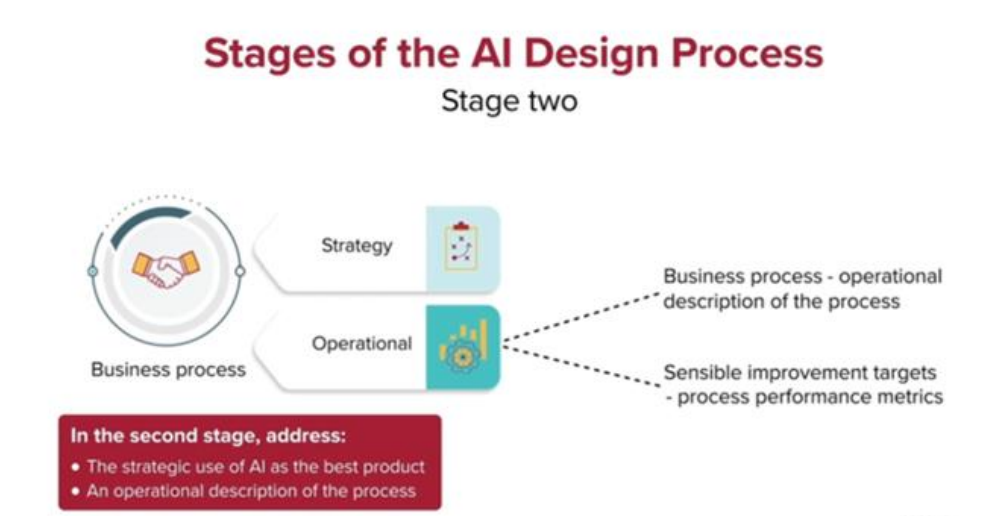




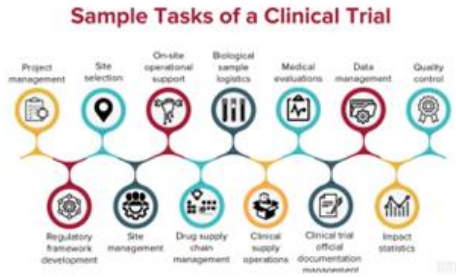


Strategy	Key Focus	Relevance to COVID
Best Product	Product Superiority	Important, but speed to market is more critical in the short term.
Network Externalities	Scaling and Lock-in	First to market, gathers most data, improves rapidly due to time pressure.
Full Customer Solutions	Integrated Medical Solutions	Can be pursued later; initial focus on scaling diagnostic capabilities.

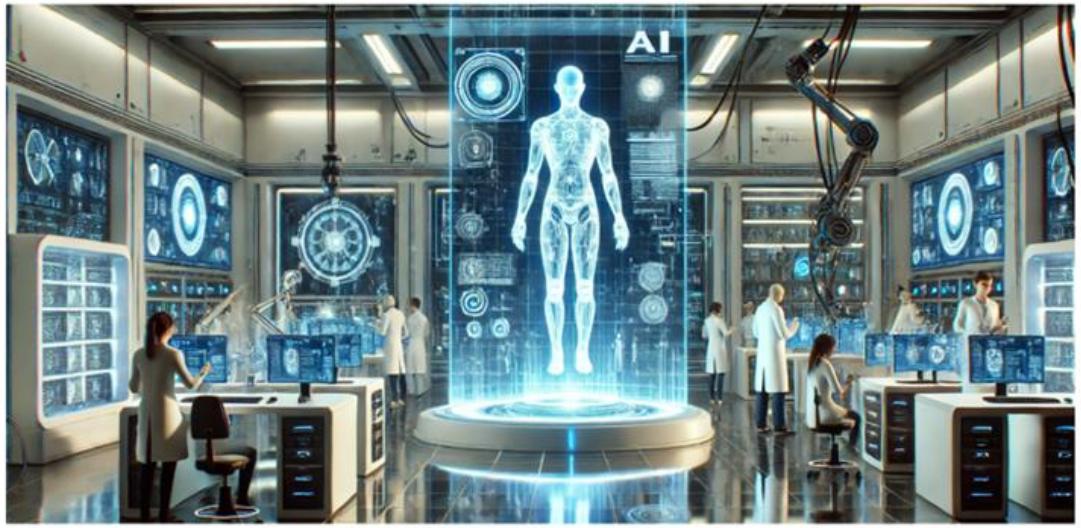
Stage II: Operational Decision — Alzheimer's







- AI can help develop biomarkers to track disease progression and drug effectiveness in clinical trials
- AI optimizes and streamlines trial management, reducing time and costs by enhancing regulatory, medical, and data processes.
- FDA and health officials must approve AI tools, requiring significant coordination and resources.
- AI's impact on operations affects project management and requires alignment from doctors, which involves considerable operational costs.
- Success depends on investing in both AI and necessary support processes, beyond just development focusing on strategic resource allocation



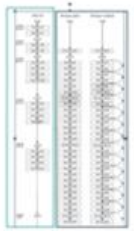
A block of ResNet:

- Adding layers to a simple convolutional network produces a phenomenon whereby adding layers decreases the system performance.
- This issue was addressed through a signal that's pushed forward, thereby training the network on residual differences.

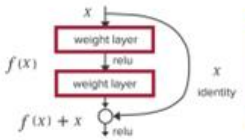


Layers of Convolutional Network

• Example of ResNet



• A block of ResNet



Addresses a particular problem of convolutional networks called the vanishing gradient problem



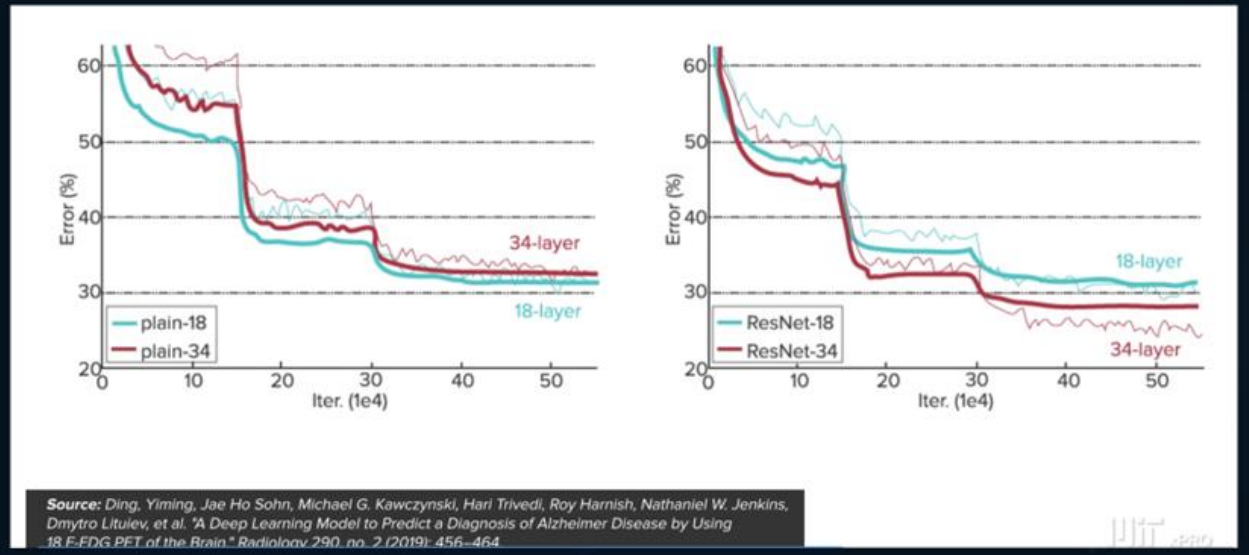
AI Programming Package

TensorFlow



IP Approach: The Choice of ResNets





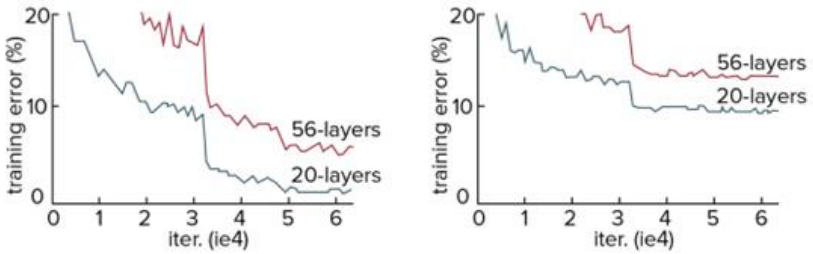
Source: *Ding, Yiming, Jae Ho Sohn, Michael G. Kawczynski, Hari Trivedi, Roy Harnish, Nathaniel W. Jenkins, Dmytro Lituiev, et al. "A Deep Learning Model to Predict a Diagnosis of Alzheimer Disease by Using 18 F-FDG PET of the Brain." Radiology 290, no. 2 (2019): 456–464.*

Understanding the Base Case in Deep Learning Models

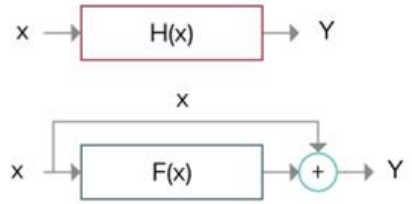
- Start with existing models before developing new architectures which results in a base case comparison
- More layers don't always improve performance, as shown by the 18-layer network outperforming the 34-layer one in CNNs
- Deeper networks (like ResNet-34) can outperform smaller ones when designed correctly in this case a ResNet advantage
- Use base cases to benchmark and guide future model enhancements resulting in iterative improvements
- When developing your AI product strategy, establish a base case to compare product iterations and competitor models effectively

Key Concepts of ResNets

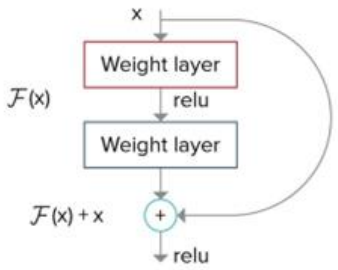
1. Handling the degradation problem



2. Residual learning



3. Skip connections



x
identity

Handling the degradation problem

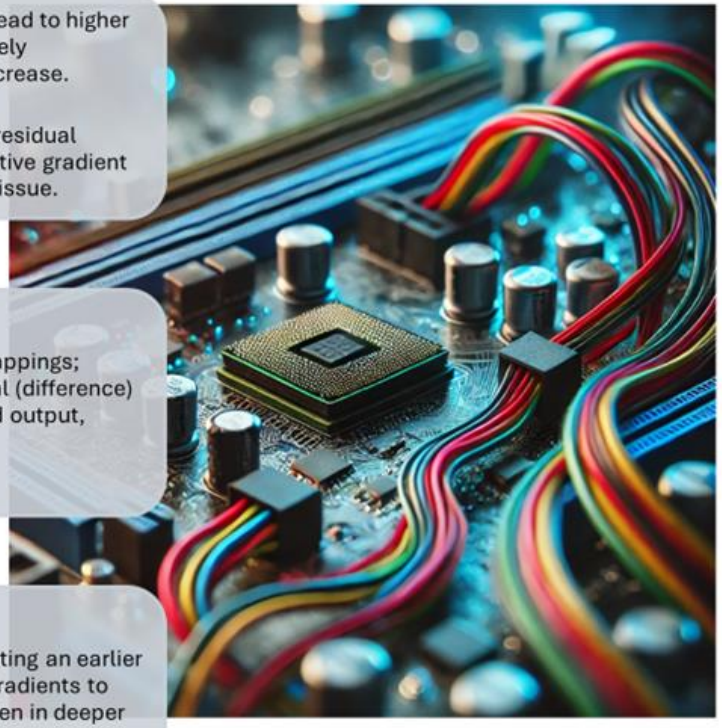
- Issue: Deeper networks can lead to higher training errors, counterintuitively performing worse as layers increase.
- Solution: ResNets introduce residual connections that enable effective gradient flow, solving this degradation issue.

Residual learning

- ResNets don't learn direct mappings; instead, they learn the residual (difference) between the input and desired output, which is easier to optimize.

Skip connections

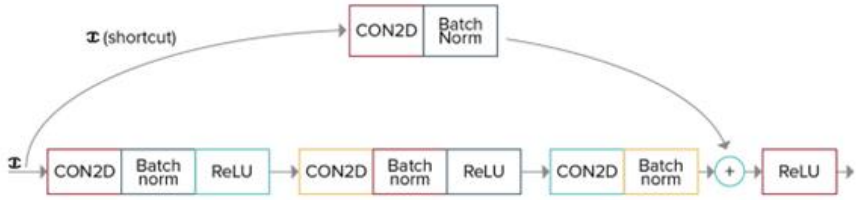
- Layers are skipped by connecting an earlier layer to a later one, allowing gradients to propagate more efficiently, even in deeper networks.



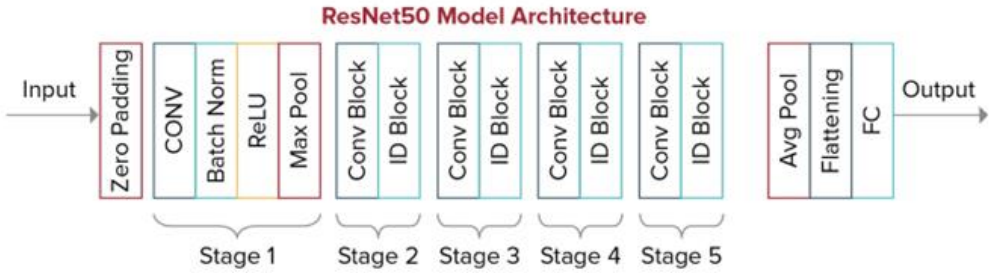
He, K., Zhang, X., Ren, S., and Sun, J. (2015) Deep residual learning for image recognition Links to an external site.. Microsoft Research.

Key Concepts of ResNets

4. Building blocks



5. ResNet variants



Building Blocks

- A typical residual block consists of multiple layers (convolution, batch normalization, and ReLU) with a shortcut connection bypassing these layers.

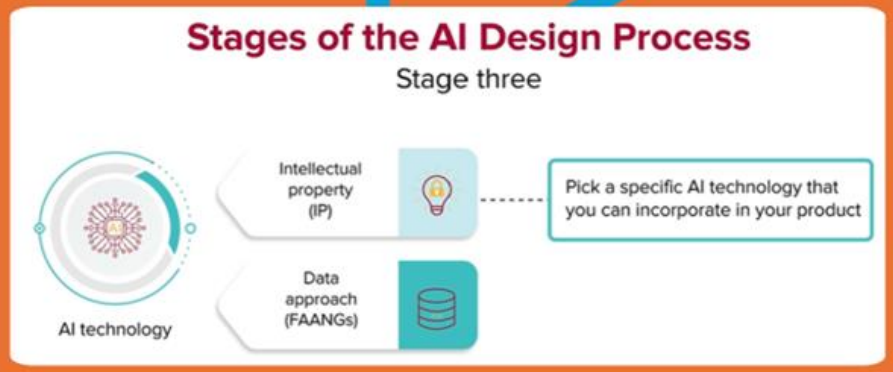
ResNet variants

- Different versions exist based on layer depth, such as ResNet18, ResNet50, and ResNet101, providing varying levels of model complexity and performance.

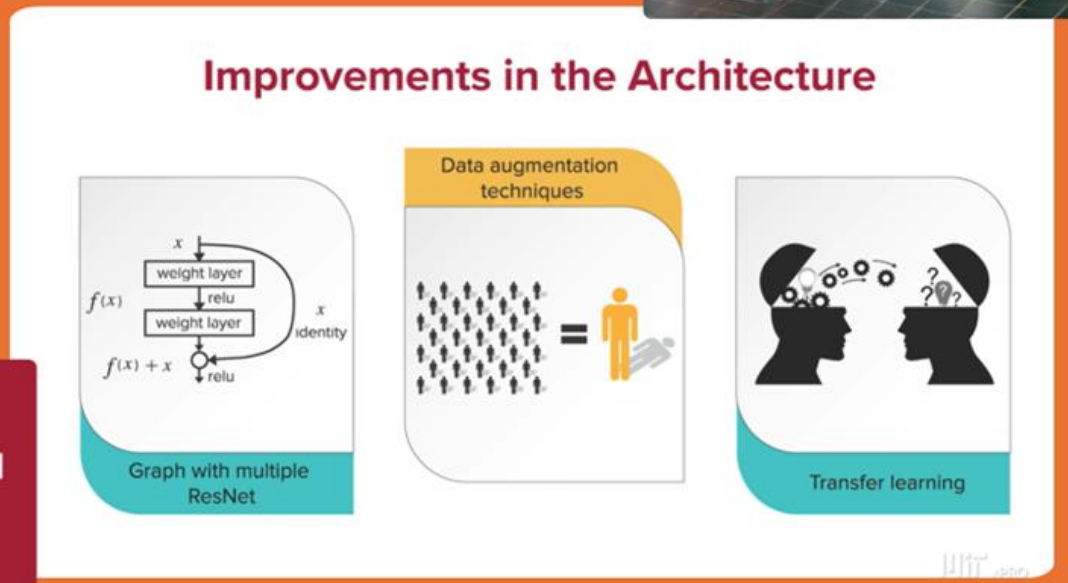


He, K., Zhang, X., Ren, S., and Sun, J. (2015) Deep residual learning for image recognition Links to an external site.. Microsoft Research.

Stage III — (Part 1: IP) The Choice of the Architecture



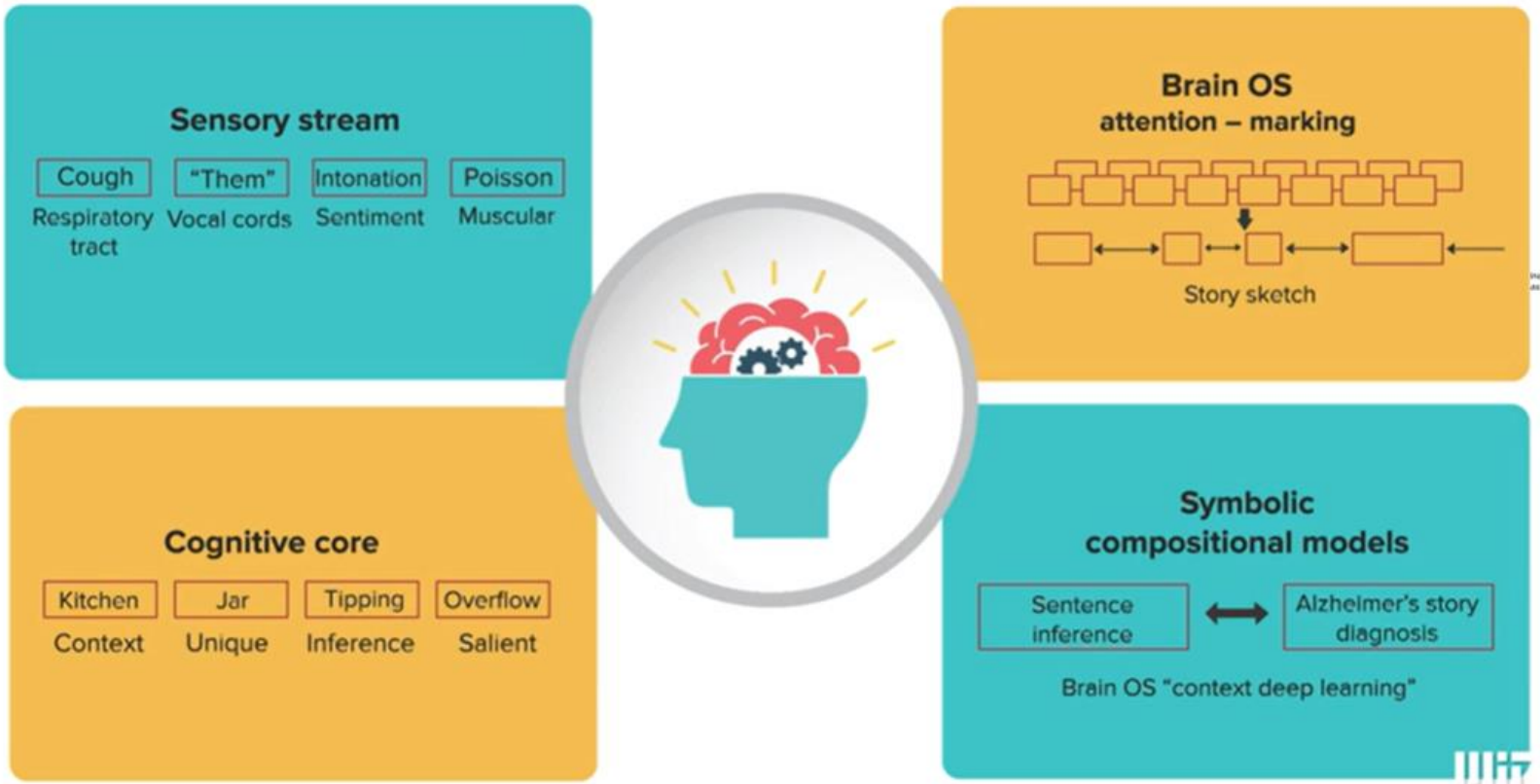


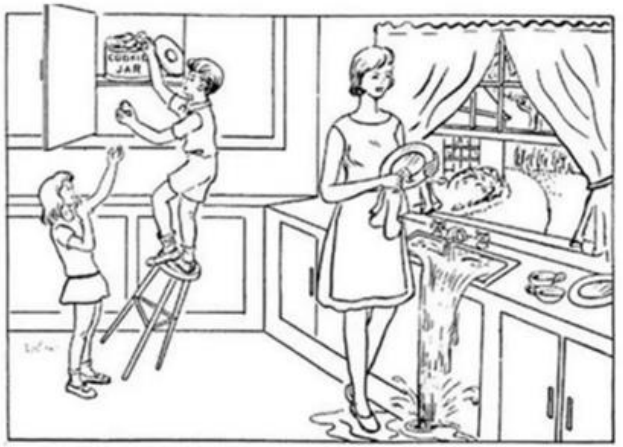


Open-Voice Brain Model architecture (OVBM):

- Identify and combine biomarkers
- Find malfunctioning components of the brain

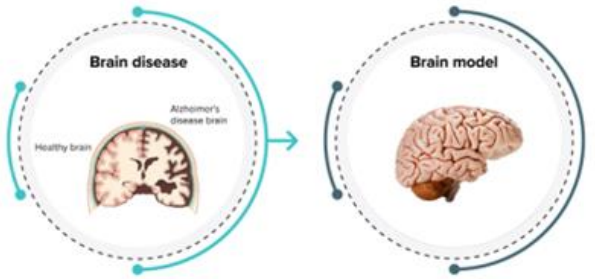
MIT Open-Voice Brain Model





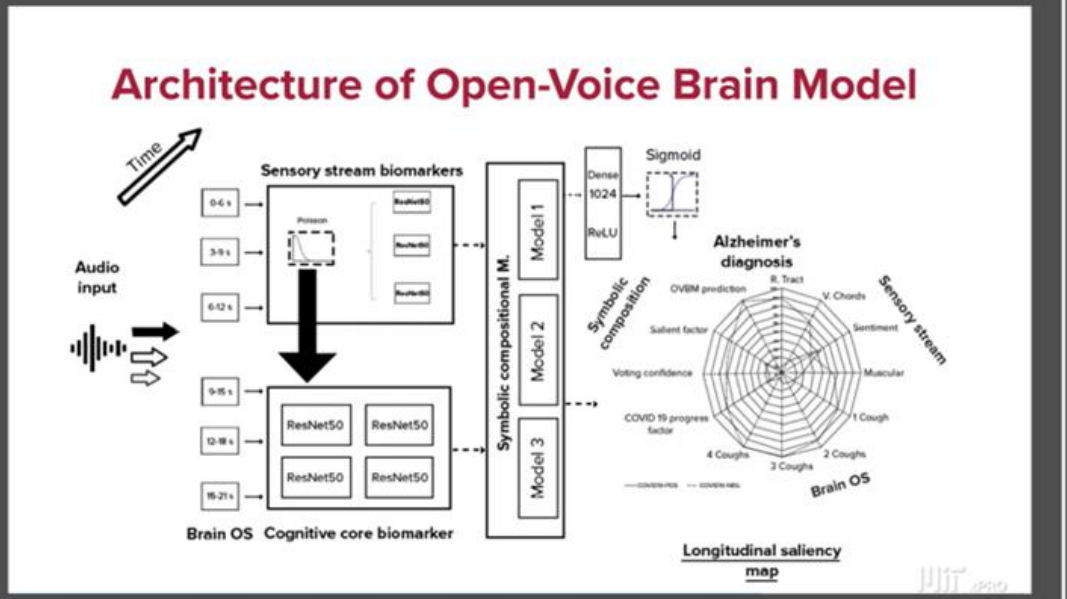
Inakis, Dimitrios, Kristina Lundholm Fors, Eva Björkner, and Arto Nordlund. "Data Collection from Persons with Mild Forms of Cognitive Impairment and Healthy Controls for Classification and Prediction of Dementia. 2017.

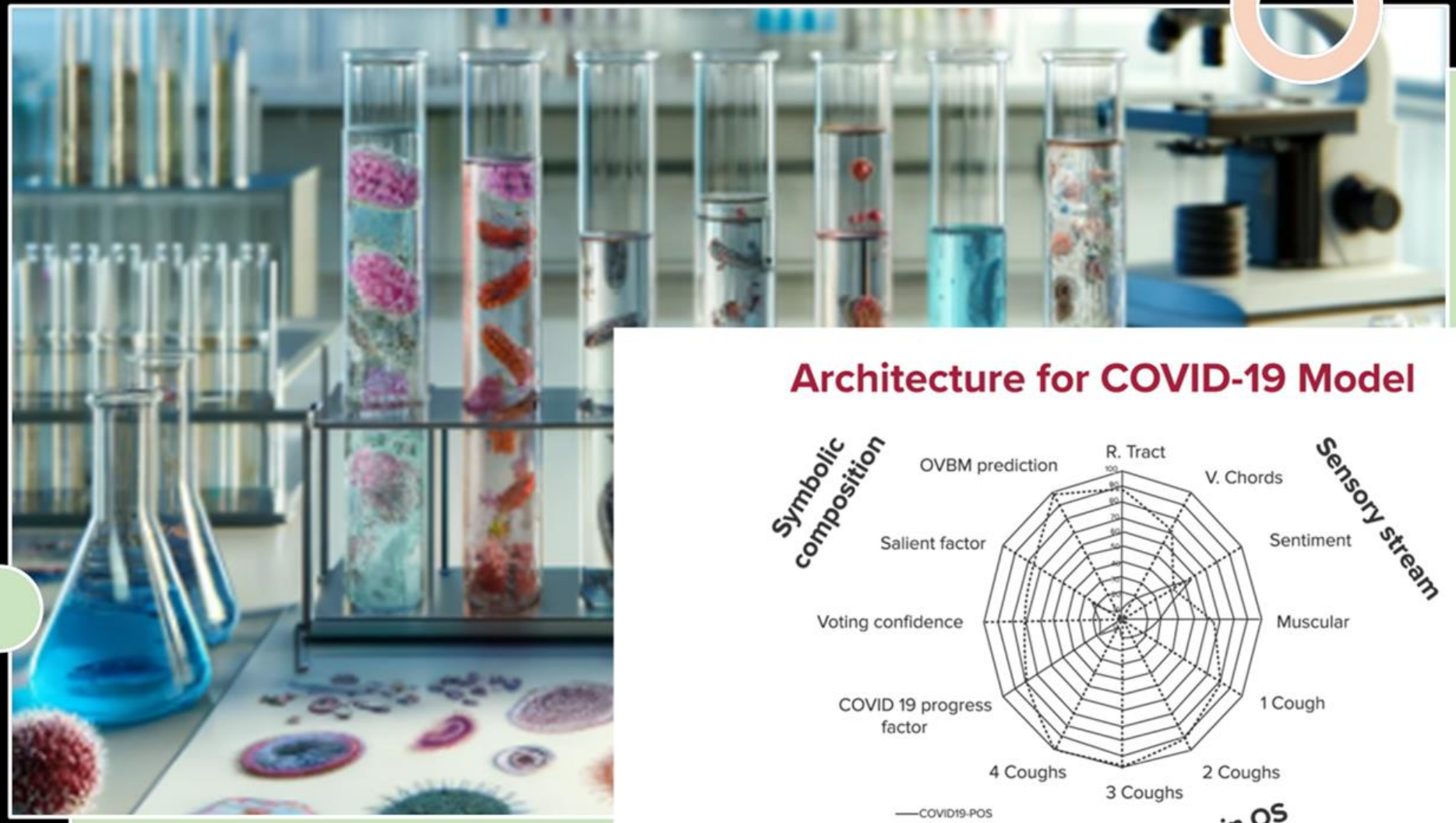
A Reference Model for Alzheimer's Disease



Architecture for Cognitive Biomarker Mapping in Alzheimer's Detection




Architecture of Open-Voice Brain Model
Time
Audio input
Sensory stream biomarkers
Poisson
0-6 s
3-9 s
6-12 s
9-15 s
12-18 s
15-21 s
ResNet50
ResNet50
ResNet50
ResNet50
Brain OS
Cognitive core biomarker
Symbolic compositional M.
Model 1
Model 2
Model 3
Dense 1024
ReLU
Sigmoid
Alzheimer's diagnosis
Symbolic composition
Sensory stream
OVBM prediction
R. Tract
V. Chords
Sentiment
Muscular
1 Cough
2 Coughs
3 Coughs
4 Coughs
COVID 19 progress factor
Voting confidence
Salient factor
Brain OS
Longitudinal saliency map



Architecture for COVID-19 Model

Self-Study Discussion 4.1: How Do You Augment a Training Database? [30 minutes]

Learning Outcome Addressed

4. Describe how to augment a dataset using biomarkers.

This is a self-study discussion and does not count toward the completion of the program.

Introduction

The research paper COVID-19 Artificial Intelligence Diagnosis Using Only Cough Recordings (Lauguarta et al., 2020) presents the dataset, model architecture, and results of a zero-cost, rapid COVID-19 forced-cough recording AI prescreening tool. The paper summarizes four biomarkers that the authors used to improve and augment the training database:

- **Biomarker 1** (Muscular degradation): Following memory decay, models from the authors introduced muscle fatigue and degradation features by modifying input signals for all train and test sets with the Poisson mask in the model.
- **Biomarker 2** (Vocal cords): Subjects with lung diseases often have distinct expressions of vocal cord biomarkers compared to healthy ones.
- **Biomarker 3** (Sentiment): Clinical evidence supports the presence of cognitive decline and emotional distress in COVID-19 patients.
- **Biomarker 4** (Lungs and respiratory tract): Respiratory infections alter the physical structure of the lungs and respiratory tract. In the early days of COVID-19, epidemiologists listened to the lungs while patients forced coughing as part of the diagnostic methods.

Task

In this exercise, you will focus on biomarkers 2 (vocal cords) and 3 (sentiment). In the previous activities, you may have found it difficult to distinguish between the types of cough by listening to them. For this reason, AI can become useful when augmenting a dataset by distinguishing between coughs.

In your post, address the following questions:

1. How would you use the different emotions and sounds to augment your dataset?
2. What types of metrics would you define to augment your dataset?

Read the statements posted by your peers. Feel free to respond to your peers with thoughtful comments and questions to deepen the discussion.

Submission Instructions

- The suggested word count for this discussion is 100-150 words.
- The average completion time for this activity is 30 minutes.

Stage III — (Part 2: Data Strategy) Privacy and IRB Approval

Stages of the AI Design Process

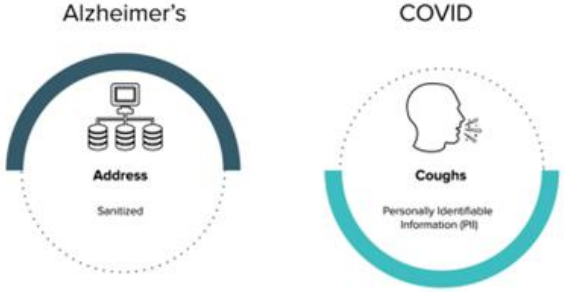
Stage three



FAANGs



Addressing Privacy in Alzheimer's and COVID Cases



General Data Protection Regulation (GDPR)

- **Internal revenue board (IRB):**
  - Specialized bodies within hospitals or universities, before you can start collecting data about humans
- **Committee on the use of humans as experimental subjects (CUHE):**

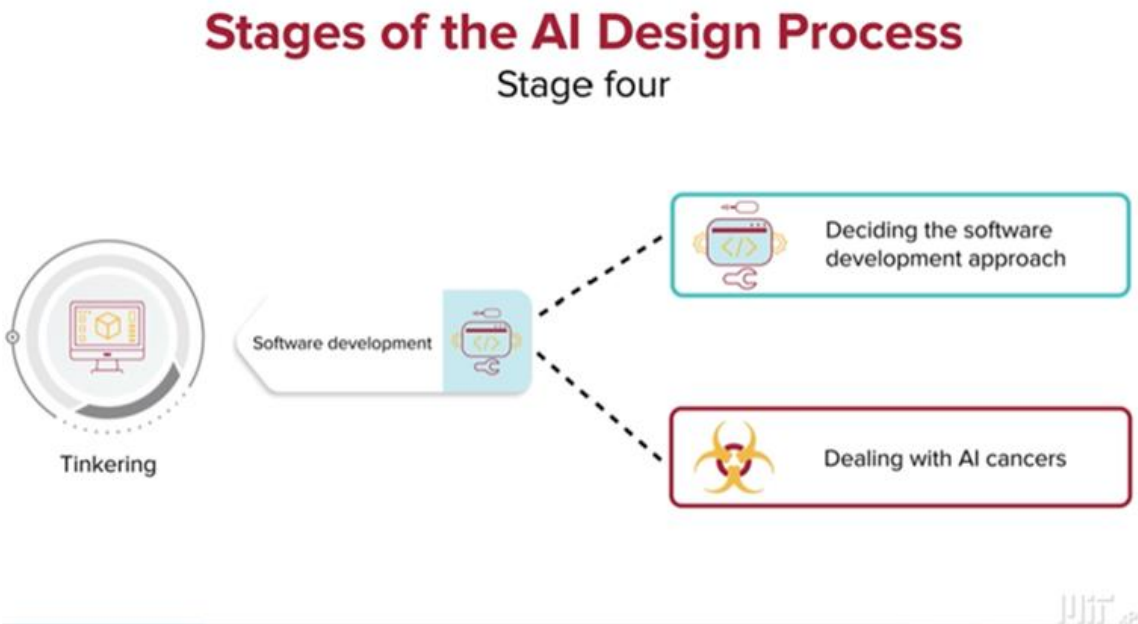


- STUDY PROCEDURES: You will submit cough data using website audio recordings that will start once you give your consent to participate by clicking next and answering six questions about your (1) age, (2) sex, (3) zip code, (4) symptoms (fever, fatigue, sore throat, difficulty breathing, persistent pain or pressure in the chest, diarrhea, cough), and whether you've undergone a COVID-19 test and what the outcome was. The total participation time should be under 30 seconds. You will record your cough using the recording feature in the website and the audio will be sent to us by the browser.



- PURPOSE: In this study, we will collect sample coughs of COVID-19 patients and that of non-patients and try to distinguish the two groups using artificial intelligence.

Stage IV — Tinkering — Software Methodology





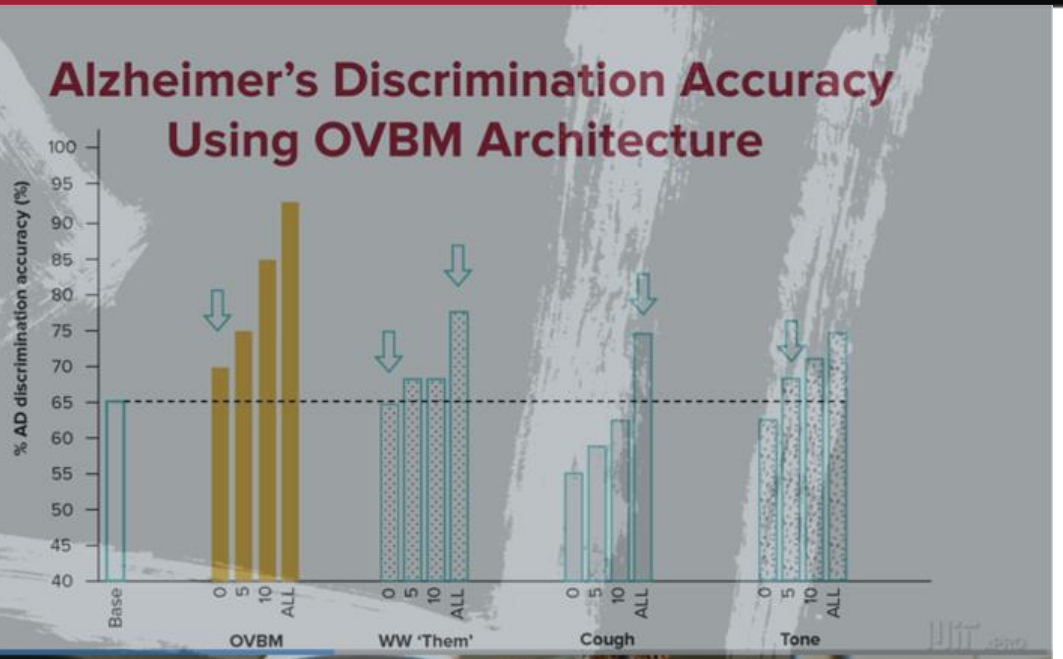
Software methodology

- Code
  - TensorFlow
  - GitHub
- Graphics Processing Unit (GPU)
  - Azure cloud with NVIDIA Tesla machines



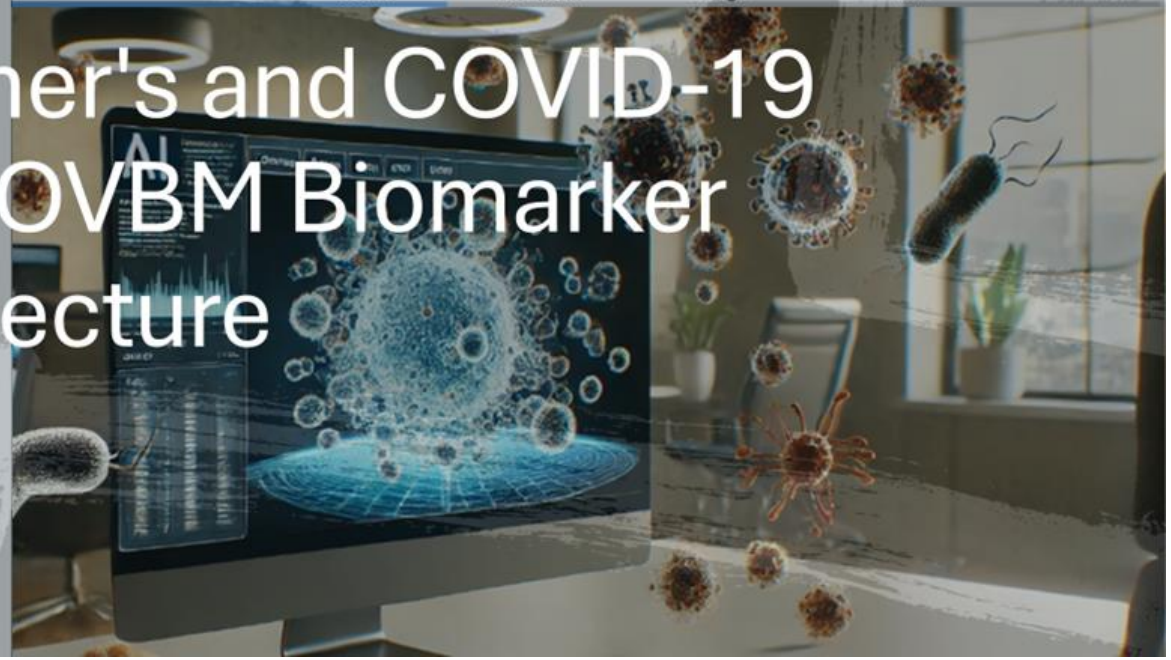
Percentage of COVID and Alzheimer's Subjects Identified by Biomarkers

Biomarker	Model Name	COVID (%)	Alzheimer (%)
R. Tract	Cough	23	9
Sentiment	Intonation	8	19
Vocal Cords	WW 'THEM'	19	16
R. Tract & Sent.	Cough & Tone	0	0
R. Tract & V. Cords	Cough & WW	1	6
Sent & V. Cords	Tone & WW	0	3
In all 3		34	41





Optimizing Alzheimer's and COVID-19 Detection Using OVBM Biomarker Architecture



Self-Study Discussion 4.2: What Are the Ingredients You Need in a Consent Form? [30 minutes]

Learning Outcome Addressed

5. Discuss the approvals necessary for human experimentation.

This is a self-study discussion and does not count toward the completion of the program.

Introduction

The MIT Committee on the Use of Humans as Experimental Subjects (COUHES) is responsible for reviewing research projects that utilize humans as research subjects and devising procedures to ensure that subjects are protected against risk and that their rights, privileges, and privacy are protected. One of the nonbiomedical research projects at MIT that utilize humans as a research subject is conducted by Professor Subirana. This project uses an AI prescreening tool to detect COVID-19 from the sound of a cough.

The following web pages will help in this activity:

- Guidelines for Selected Procedures and Populations
- Forms and Templates
- Artificial Intelligence Model Detects Asymptomatic Covid-19 Infections Through Cellphone-Recorded Coughs
- COVID-19 Open Research Dataset
- MIT Open Access Articles

Task

In your post, address the following prompts:

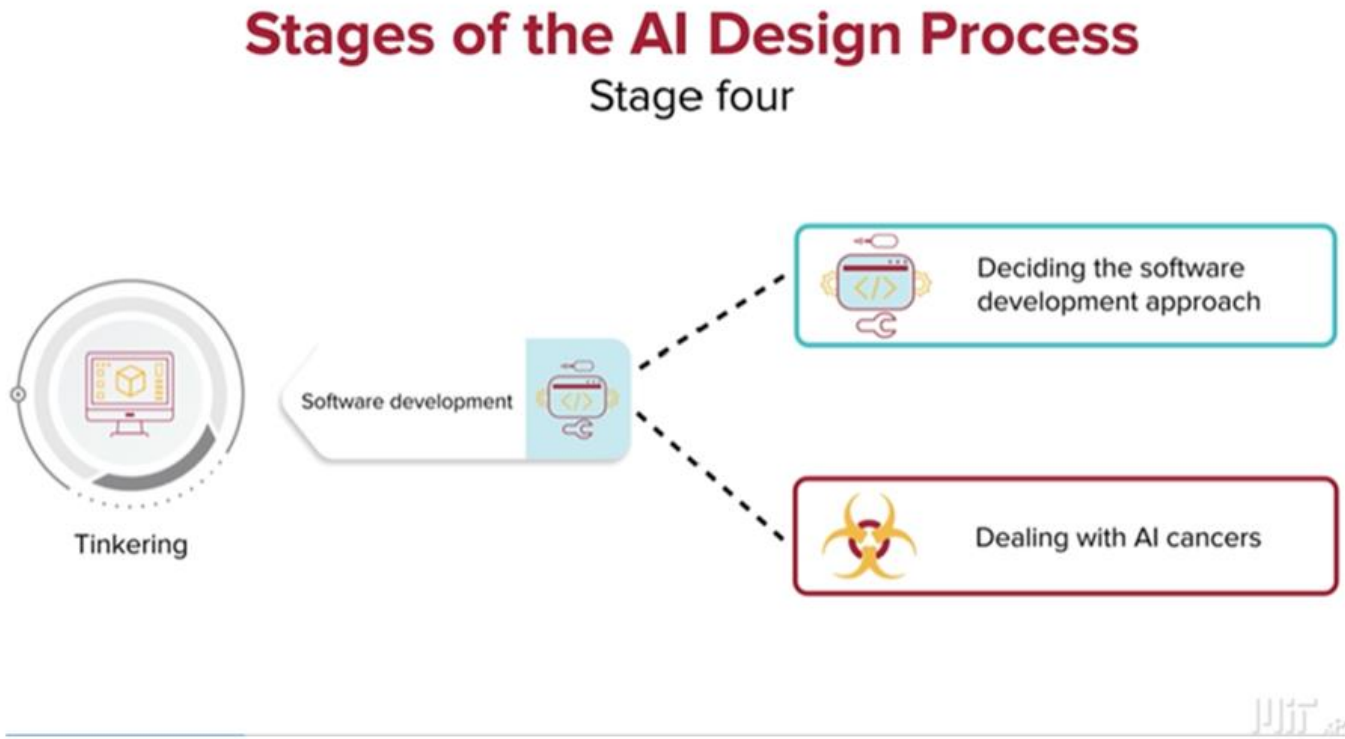
1. Discuss the fundamental parts that a consent form for this type of experiment should contain to be approved by COUHES.
2. Locate the correct template for the consent form for the desired type of project.
3. Summarize the fundamental parts that a consent form should contain to be approved.

Read the statements posted by your peers. Feel free to respond to your peers with thoughtful comments and questions to deepen the discussion.

Submission Instructions

- The suggested word count for this discussion is 150-200 words.
- The average completion time for this activity is 30 minutes.

Stage IV — Tinkering: AI Cancers



Concerns with Artificial Intelligence

- Explainability
  - Diseases have multiple causes and symptoms
- Bias
  - Biomarkers have gender bias and cultural bias

Learning Outcome Addressed

6. Discuss the implications of crowdsourcing data strategies.

This is a self-study activity and does not count toward the completion of the program.

Introduction

Imagine a researcher extracting data from a published survey. It can be challenging to find volunteers to complete a survey. In addition, AI technology cannot help with tasks such as filling out surveys and categorizing information.

Because crowdsourcing data can be challenging and time-consuming, some organizations provide services to help smaller projects and startups outsource their processes.

One example is Amazon Mechanical Turk (MTurk), a crowdsourcing marketplace for individuals and businesses to outsource their processes and jobs to a distributed workforce that can perform these tasks virtually. Tasks include conducting simple data validation and research or more subjective tasks such as survey participation and content moderation.

For example, MTurk was initially used for tasks such as categorizing information and translating audio to text. These tasks are challenging for AI. For example, Amazon uses the MTurk platform to enable potential shoppers to take a picture of a product and receive a link to the same product on Amazon. In reality, a human being is paid to match the shopper's image with an Amazon product.

Task

Find an article, a podcast, a video, or another form of media about Amazon Turk as a data source.

In your post, address the following prompts:

- What would be the pros and cons of working with MTurk rather than merely advertising your project?
- Include links to your sources.

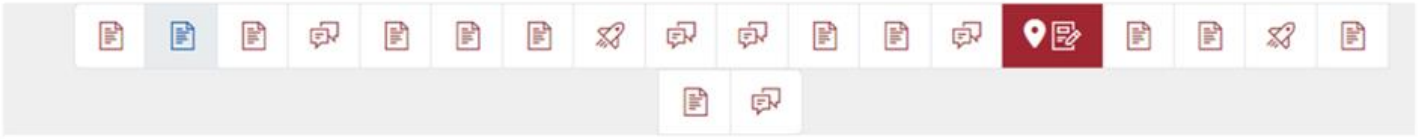
Read the statements posted by your peers. Feel free to respond to your peers with thoughtful comments and questions to deepen the discussion.

Submission Instructions

- The suggested word count for this discussion is 150-200 words.
- The average completion time for this activity is 30 minutes.

This topic was edited by Sara Huter

Solving Healthcare Problems [Jupyter Notebook 4.1]



Learning Outcome Addressed

7. Demonstrate how AI can solve healthcare problems.

This is a self-study activity and does not count toward the completion of the program.

Task

You have learned how AI can be applied to detect COVID-19 and diagnose Alzheimer's disease. In this exercise, you demonstrate how AI can do so in Python.

Submission Instructions

The average completion time for this activity is 30 minutes.

- *After opening the activity in Codio, complete and run the code following the instructions in the Jupyter Notebook Activity.*
- *Important: There is nothing to submit in Codio or Canvas for this practice assignment.*
- *Once you've tried the activity, you can view the solution in Google Colab. Find a guide to viewing your solution in Google Colab here.*
  - *Download the solution here: Self-Study Jupyter Notebook 4.1: Solving Healthcare Problems*

This tool needs to be loaded in a new browser window

Load Solving Healthcare Problems [Jupyter Notebook 4.1] in a new window


Questions?